Modulo hashing

A simple integer hash-table index is key modulo bucket_count. With 13 buckets numbered 0 through 12, key 42 maps to index 3, because 42 = 3 * 13 + 3. A collision means two keys map to the same bucket.

Separate chaining

Separate chaining keeps a collection of entries at each bucket. Keys that collide are stored together in that bucket's chain rather than occupying alternative positions in the main table. A linked list can represent a chain.

Linear probing

Linear probing handles a collision by checking consecutive table positions: h(key), (h(key) + 1) % capacity, then (h(key) + 2) % capacity, and so on. It wraps at the end and can form long primary clusters.

Tombstone deletion

In an open-addressed hash table, mark a removed entry with a tombstone. A search continues past tombstones, since another colliding key may lie farther along the probe sequence. Treating that slot as never used could incorrectly stop the search early.

Load factor and resizing

Hash-table load factor is stored_entries divided by bucket_count. A table with 15 entries and 20 buckets has load factor 0.75. When resizing changes the bucket count, recompute indices for the entries; their old bucket positions generally cannot be kept unchanged.